



Gap Analysis, New Feature Proposals & 5-Member Team Task Division

A structured audit of the current codebase (Next.js 16 frontend, Node/Express/TypeScript backend, PostgreSQL + MongoDB + Redis), followed by prioritized feature recommendations and an actionable work split across a 5-person team (M1–M5).

Project	EdLearn — AI-Powered Learning Platform
Prepared for	Ed Compass (theedcompass@gmail.com)
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Scope	Backend, Frontend, Infra/DevOps, Security, Product

1. Current Architecture Snapshot

EdLearn is a decoupled full-stack platform: a Next.js 16 (App Router) frontend with Zustand state and jsPDF exports, and a Node/Express/TypeScript backend handling RAG-based note generation, TTS audio, web scraping, and BullMQ background jobs. Data is split across PostgreSQL (Prisma — users, roadmaps, vision boards, subscriptions), MongoDB (discussion forum, market-demand trends), and Redis (caching, session/roadmap persistence, job queues). The system already includes JWT auth, rate limiting, SSRF guards, IDOR fixes, and a Kubernetes manifest set for the frontend/backend deployments.

Confirmed in place: 7 pedagogical learning modes, Duo Podcast TTS, branded PDF export, discussion board with badges, market-demand scrapers, Vision Board, SSO handoff between sibling apps (EdMentor/EdCompass/EdQuiz), and core security hardening (Sprint 2–3 log).

2. Identified Gaps

Testing & CI/CD

- No automated test suite anywhere in the repo (no Jest/Vitest, no *.test.* or *.spec.* files) — zero unit, integration, or e2e coverage.
- No CI/CD pipeline (no GitHub Actions or equivalent) — builds, lint, and Docker images are not automated.
- No load/performance testing for AI generation endpoints or scrapers.

Security & Auth

- No forgot-password / email-based password reset flow — only an authenticated change-password endpoint exists.
- No 2FA/MFA option for accounts.
- No visible refresh-token rotation — JWT session lifecycle beyond login is unclear.
- Rate limiting only covers auth and AI routes; books/media/news/vision-board routes appear unprotected.
- No centralized input-validation layer (e.g. zod/yup) — validation is done ad hoc (manual clamping in a couple of places).
- No GDPR-style data export or account-deletion flow for users.

Infrastructure & DevOps

- TTS audio files are written to local disk (backend/tts-audio/) — not durable, won't survive redeployments, and won't scale across the 2+ replica K8s deployment.
- K8s manifests define only ConfigMaps, no Secrets objects — credentials handling in-cluster is unclear.
- No Horizontal Pod Autoscaler (HPA) or resource limits visible in the K8s deployment specs.
- No monitoring/observability stack (no Sentry, Prometheus/Grafana, or structured logging framework like Winston/Pino).
- No documented backup/disaster-recovery strategy for PostgreSQL or MongoDB volumes.
- No automated Docker image build/push step tied to CI.

Product & Backend Completeness

- Sprint 4 roadmap items are still open: full BullMQ worker rollout, Edge TTS / ElevenLabs upgrade (currently on free google-tts-api).
- Subscription model exists in the Prisma schema but no Stripe (or other payment provider) routes/webhooks were found — monetization is not wired up.
- No full-text or semantic search across generated notes, roadmaps, or discussion threads.
- No moderation layer for AI-generated content or the scraped news/media/books feeds.
- Single user role only — no mentor/teacher/parent role or classroom/cohort model.

Frontend/UX & Housekeeping

- No accessibility (WCAG) audit has been performed on the frontend.
- No internationalization (i18n) — UI and generated notes are English-only.
- No offline support or PWA manifest for the study workspace.
- Discussion board is REST-only — no real-time updates (WebSockets) for new answers/upvotes.
- Dead code present (e.g. components/Landing/_to_delete/) and a near-empty, seemingly stray root package.json that should be cleaned up or clarified.
- No email notification system (reminders, digests, streak alerts).

3. Proposed Additional Features

Prioritized by impact on growth, retention, and platform completeness.

#	Feature	Why it matters
1	Stripe subscription & billing	Activates the existing Subscription model; enables paid tiers and credit top-ups.
2	Real-time discussion board (WebSockets)	Live answers/upvotes instead of poll-on-refresh; higher engagement.
3	AI-generated quizzes & spaced-repetition flashcards	Reinforces learning modes with active-recall practice.
4	Live conversational AI voice tutor	Upgrades passive Duo Podcast into an interactive spoken Q&A session.
5	Gamification: streaks, XP, leaderboards	Extends the existing badge system into daily-habit retention loops.
6	Email/notification system	Study reminders, streak-at-risk alerts, weekly progress digests.
7	Admin & moderation dashboard	Review scraped content, manage users, monitor AI generation costs.
8	Full-text / vector search	Search across notes, roadmaps, and forum threads instantly.
9	Collaborative study groups	Shared roadmaps and group discussion boards for cohort learning.
10	PWA / offline mode	Cached notes and offline-first workspace for mobile users.
11	Calendar sync (Google Calendar)	Pushes daily study blocks straight into the user's calendar.
12	Analytics dashboard for learners	Time-on-topic, mastery heatmaps, retention curves.
13	Additional OAuth providers	Microsoft & GitHub sign-in alongside existing Google OAuth.
14	Notion/Slack export	One-click export of study notes into tools students already use.
15	Multi-language (i18n) notes & UI	Opens the platform to non-English-speaking learners.

4. Task Division — 5 Team Members (M1–M5)

Work is grouped into five balanced workstreams combining gap fixes and new features. Each member owns an end-to-end domain to minimize cross-dependencies.

M1 — Backend Core, Security & Testing

- Set up Jest/Vitest + Supertest; write unit & integration tests for auth, roadmap, and generate routes.
- Build GitHub Actions CI: lint, type-check, test, Docker build on every PR.
- Add zod-based request validation across all backend routes.
- Implement forgot-password (email OTP/link) flow and refresh-token rotation.
- Add 2FA (TOTP) as an opt-in account setting.
- Extend rate limiting to books/media/news/vision-board routes.
- Add GDPR data export & account deletion endpoints.

M2 — Infrastructure & DevOps

- Migrate TTS/media file storage from local disk to S3/Cloudflare R2.
- Add Kubernetes Secrets objects and resource requests/limits to all deployments.
- Configure Horizontal Pod Autoscaler (HPA) for backend/frontend.
- Stand up monitoring: Sentry for error tracking, Prometheus + Grafana for metrics, structured logging (Pino).
- Document and script Postgres/Mongo backup & restore procedure.
- Wire Docker image build/push into the CI pipeline from M1.

M3 — AI, Content & Search

- Complete Sprint 4: full BullMQ worker coverage for scrapers and TTS jobs.
- Upgrade TTS to Edge TTS / ElevenLabs for Duo Podcast mode.
- Build live conversational AI voice tutor (streaming STT + LLM + TTS).
- Add AI-generated quizzes and spaced-repetition flashcards per topic.
- Implement full-text/vector search across notes, roadmaps, and forum threads.
- Add a lightweight moderation pass for scraped news/media/books content.

M4 — Monetization, Admin & Growth

- Integrate Stripe: checkout, webhooks, tier gating, and credit top-ups against the existing Subscription model.
- Build an admin/moderator dashboard (users, content review, AI cost monitoring).
- Build a learner analytics dashboard (time-on-topic, mastery heatmaps, retention).
- Implement email notification system (reminders, streak alerts, weekly digest).
- Add gamification: streaks, XP levels, leaderboards on top of the existing badge system.
- Add a referral/affiliate program.

M5 — Frontend, UX & Platform Reach

- Run a WCAG 2.1 AA accessibility audit and fix the top issues (contrast, keyboard nav, focus states).
- Add PWA support (manifest, service worker, offline note caching).
- Build real-time discussion board updates via WebSockets/Socket.io.

- Add i18n scaffolding for UI and generated notes (starting with 1–2 additional languages).
- Build collaborative study groups (shared roadmaps + group discussion threads).
- Add Google Calendar sync for daily study blocks; add Microsoft/GitHub OAuth.
- Clean up dead code (`_to_delete` folder, stray root `package.json`) and add a dark/light theme toggle.

5. Suggested Delivery Timeline

A 4-phase, ~8-week rollout runnable in parallel across M1–M5.

Phase	Weeks	Focus	Owners
1. Foundations	1–2	Testing setup, CI/CD, secrets/storage migration, validation layer	M1, M2
2. Core Upgrades	3–4	BullMQ completion, TTS upgrade, Stripe integration, password reset	M1, M2, M4
3. Engagement Features	5–6	Quizzes/flashcards, gamification, real-time forum, admin dashboard	M3, M4, M5
4. Reach & Polish	7–8	Accessibility, PWA/offline, i18n, calendar sync, monitoring go-live	M2, M5

Note: workstreams are designed to run largely in parallel; M1's CI/testing setup and M2's secrets/storage work should land first since other members' features depend on a stable pipeline and durable file storage.